

ϵ -Minimax Guarantees under Squint

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Abstract

In this short note, we seek to answer the following question: can the use of the Hedge algorithm for getting ϵ -minimax solutions in [Aradillas Fernández et al. \(2025\)](#) be replaced with the Squint algorithm to get better theoretical guarantees? The answer we find is, in general, no. The theoretical guarantees in Theorem 1 of [Aradillas Fernández et al. \(2025\)](#) are stronger than what the Squint algorithm grants in the worst-case. However, if cumulative squared regret can be bounded better than the worst-case of M^2T , then it is possible to get a better theoretical guarantee than in [Aradillas Fernández et al. \(2025\)](#), but this improvement would have to be problem-specific, and runs against the agnostic nature of Hedge in [Aradillas Fernández et al. \(2025\)](#).

1 Second-Order Minimax Optimization via Squint

We consider the minimax problem introduced in [Aradillas Fernández et al. \(2025\)](#). Recall that $\mathcal{D} = \{d_1, \dots, d_I\}$ is a finite collection of decision rules, $\Delta(\mathcal{D})$ denotes the simplex over these rules, and

$$R(p, \theta) = \sum_{i=1}^I p_i R(d_i, \theta), \quad f(p) = \sup_{\theta \in \Theta} R(p, \theta), \quad \bar{v} = \inf_{p \in \Delta(\mathcal{D})} f(p).$$

Under Assumption A1¹, $0 \leq R(d_i, \theta) \leq M$ for every i and θ . Under Assumption A2, for every $p \in \Delta(\mathcal{D})$ there exists a maximizer θ_p satisfying $R(p, \theta_p) = f(p)$. In Algorithm A1, at iteration t the statistician selects $p_t \in \Delta(\mathcal{D})$, nature chooses a best response

$$\theta_t \in \arg \max_{\theta \in \Theta} \sum_{i=1}^I p_{i,t} R(d_i, \theta),$$

and the resulting risk vector is

$$g_t = (R(d_1, \theta_t), \dots, R(d_I, \theta_t))^\top.$$

Consequently,

$$p_t^\top g_t = f(p_t).$$

The structure of this problem permits us to replace the Hedge update in Algorithm A1 by the Squint algorithm of [Koolen and Van Erven \(2015\)](#), which essentially consists of sticking in a second-order term into the weight update. To see this, define the normalized losses

$$\ell_{i,t} \equiv \frac{g_{i,t}}{M} = \frac{R(d_i, \theta_t)}{M} \in [0, 1]$$

and the normalized loss of the statistician at iteration t by

$$\hat{\ell}_t \equiv p_t^\top \ell_t = \sum_{i=1}^I p_{i,t} \ell_{i,t}.$$

The instantaneous regret relative to decision rule d_i is then

$$r_{i,t} \equiv \hat{\ell}_t - \ell_{i,t}.$$

¹This is how we will enumerate assumptions/other content from [Aradillas Fernández et al. \(2025\)](#). To map to the original paper, ignore the “A” preceding the enumeration.

We define its cumulative regret and cumulative squared regret by

$$\mathcal{R}_{i,t} \equiv \sum_{s=1}^t r_{i,s}, \quad \mathcal{V}_{i,t} \equiv \sum_{s=1}^t r_{i,s}^2.$$

Thus, in the original units of the statistical decision problem,

$$\mathcal{V}_{i,t} = \frac{1}{M^2} \sum_{s=1}^t (p_s^\top g_s - g_{i,s})^2.$$

The second-order quantity entering Squint is therefore the sum of squared instantaneous regrets, rather than the square of cumulative regret. This is the term we will include in the weight-update. Intuitively, this is placing more weight on the decision rules that have loss closer to the running sum of squared regrets, i.e., we are “trusting” the algorithm to lead us to a nice solution, and so believe the “better” decision rules are ones close to the running sum in terms of loss.

The fact that the risk vector g_t depends on p_t through nature’s best response does not interfere with the Squint regret argument. The Squint potential inequality is pathwise and applies to an arbitrary sequence of loss vectors in $[0, 1]^I$. Hence the loss vector at round t may be selected after observing p_t , as occurs here when nature computes a best response.

1.1 A fixed-learning-rate second-order update

It is useful first to consider the specialization of the Squint potential to a fixed learning rate $\eta \in (0, 1/2]$, and then use the full adaptive approach for the learning rate used in [Koolen and Van Erven \(2015\)](#). In normalized units, the corresponding unnormalized weights evolve according to

$$w_{i,t} = w_{i,t-1} \exp \{ \eta r_{i,t} - \eta^2 r_{i,t}^2 \},$$

with the initial weights assumed to be uniform. Since

$$\eta r_{i,t} = \eta \widehat{\ell}_t - \eta \ell_{i,t}$$

and $\eta \widehat{\ell}_t$ is common to all i , it cancels when the weights are normalized. Hence the same probability vector is obtained from

$$w_{i,t} = w_{i,t-1} \exp \left\{ -\eta \ell_{i,t} - \eta^2 (\widehat{\ell}_t - \ell_{i,t})^2 \right\}.$$

Returning to the original risk scale and defining $\lambda = \eta/M$, this becomes

$$w_{i,t} = w_{i,t-1} \exp \left\{ -\lambda g_{i,t} - \lambda^2 (p_t^\top g_t - g_{i,t})^2 \right\}, \quad 0 < \lambda \leq \frac{1}{2M}.$$

Thus, a direct second-order analogue of line 8 of Algorithm A1 adds a negative term proportional to the squared instantaneous excess risk (i.e., swap this boxed equation in for line 8 of Algorithm A1 and we now have the Squint algorithm). Equivalently,

$$w_{i,t} \propto \exp \left\{ -\lambda \sum_{s=1}^t g_{i,s} - \lambda^2 \sum_{s=1}^t (p_s^\top g_s - g_{i,s})^2 \right\}.$$

This fixed-learning-rate update makes clear how the second-order term enters the algorithm. It is not, however, the fully adaptive Squint procedure, since full Squint aggregates simultaneously over a continuum of learning rates.

1.2 The Squint minimax algorithm

We now describe the full Squint procedure using the improper learning-rate prior of [Koolen and Van Erven \(2015\)](#). Let $\pi = (\pi_1, \dots, \pi_I)$ be a prior distribution over the decision rules. A more direct comparison to Hedge (and the previous section) can be seen if we let this prior distribution be the uniform distribution, which we do in the next section. We initialize

$$\mathcal{R}_{i,0} = \mathcal{V}_{i,0} = 0, \quad i = 1, \dots, I.$$

At iteration t , define

$$q_{i,t} \equiv \pi_i \int_0^{1/2} \exp \{ \eta \mathcal{R}_{i,t-1} - \eta^2 \mathcal{V}_{i,t-1} \} d\eta$$

and set

$$p_{i,t} = \frac{q_{i,t}}{\sum_{j=1}^I q_{j,t}}.$$

Nature then selects

$$\theta_t \in \arg \max_{\theta \in \Theta} \sum_{i=1}^I p_{i,t} R(d_i, \theta),$$

after which we compute

$$g_{i,t} = R(d_i, \theta_t), \quad \ell_{i,t} = \frac{g_{i,t}}{M}, \quad \widehat{\ell}_t = \sum_{i=1}^I p_{i,t} \ell_{i,t},$$

and update

$$r_{i,t} = \widehat{\ell}_t - \ell_{i,t}, \quad \mathcal{R}_{i,t} = \mathcal{R}_{i,t-1} + r_{i,t}, \quad \mathcal{V}_{i,t} = \mathcal{V}_{i,t-1} + r_{i,t}^2.$$

After T iterations, as in Algorithm A1, we return the averaged strategy

$$\bar{p}_T \equiv \frac{1}{T} \sum_{t=1}^T p_t.$$

The integral defining $q_{i,t}$ admits a closed-form expression in terms of the error function, so the Squint update can be evaluated separately for every i without numerical integration. However, the authors of [Koolen and Van Erven \(2015\)](#). In particular, the replacement of Hedge by Squint does not alter the nature-oracle problem that must be solved at each iteration, and so this is still a bottle-neck to implementing an ϵ -minimax algorithm.

1.3 A second-order minimax guarantee

The principal advantage of Squint in the present problem is that its convergence guarantee can depend on the realized cumulative squared regret rather than on the worst-case upper bound $M^2 T$. We state the result for the uniform prior $\pi_i = 1/I$.

Theorem 1 (Second-order minimax guarantee). *Suppose Assumptions A1–A2 hold, and run the Squint minimax procedure above for T iterations with the uniform prior $\pi_i = 1/I$. Let*

$$i_T^* \in \arg \min_{i \in \{1, \dots, I\}} \sum_{t=1}^T R(d_i, \theta_t)$$

denote the best fixed decision rule against the sequence of best responses generated by nature, and define

$$S_T^* \equiv \sum_{t=1}^T \left[\sum_{j=1}^I p_{j,t} R(d_j, \theta_t) - R(d_{i_T^*}, \theta_t) \right]^2.$$

Then

$$f(\bar{p}_T) - \bar{v} \leq \frac{1}{T} \left[\sqrt{2S_T^*} \left(1 + \sqrt{2 \log \left[I \left(\frac{1}{2} + \log(T+1) \right) \right]} \right) + 5M \log [1 + I(1 + 2 \log(T+1))] \right].$$

Proof. cf. Theorem 4 of [Koolen and Van Erven \(2015\)](#). The proof proceeds from this result and adjusts for notation and bounds on the regret function.

Because $0 \leq R(d_i, \theta) \leq M$, the normalized losses satisfy $\ell_{i,t} \in [0, 1]$. Therefore, the second-order regret guarantee for Squint with the improper prior applies to the realized sequence $\ell_1, \dots, \ell_T$ (Theorem

4 of [Koolen and Van Erven \(2015\)](#)). Applied to the singleton reference set $\{i_T^*\}$ and the uniform expert prior, this guarantee gives

$$\begin{aligned} \sum_{t=1}^T (\hat{\ell}_t - \ell_{i_T^*, t}) &\leq \sqrt{2\mathcal{V}_{i_T^*, T}} \left(1 + \sqrt{2 \log \left[I \left(\frac{1}{2} + \log(T+1) \right) \right]} \right) \\ &\quad + 5 \log [1 + I (1 + 2 \log(T+1))]. \end{aligned}$$

By definition,

$$\mathcal{V}_{i_T^*, T} = \sum_{t=1}^T (\hat{\ell}_t - \ell_{i_T^*, t})^2 = \frac{S_T^*}{M^2}.$$

Multiplying the preceding Squent regret bound by M therefore yields

$$\begin{aligned} \sum_{t=1}^T [p_t^\top g_t - g_{i_T^*, t}] &\leq \sqrt{2S_T^*} \left(1 + \sqrt{2 \log \left[I \left(\frac{1}{2} + \log(T+1) \right) \right]} \right) \\ &\quad + 5M \log [1 + I (1 + 2 \log(T+1))]. \end{aligned} \quad (1)$$

It remains to translate this online regret inequality into a minimax guarantee. By convexity of f ,

$$f(\bar{p}_T) = f\left(\frac{1}{T} \sum_{t=1}^T p_t\right) \leq \frac{1}{T} \sum_{t=1}^T f(p_t).$$

Assumption A2 and the definition of θ_t imply

$$f(p_t) = R(p_t, \theta_t) = \sum_{i=1}^I p_{i,t} R(d_i, \theta_t) = p_t^\top g_t,$$

and hence

$$f(\bar{p}_T) \leq \frac{1}{T} \sum_{t=1}^T p_t^\top g_t. \quad (2)$$

Let $p^* \in \arg \min_{p \in \Delta(\mathcal{D})} f(p)$, so that $f(p^*) = \bar{v}$. Such a minimizer exists because, under Assumption A1, f is continuous on the compact simplex $\Delta(\mathcal{D})$. By the definition of i_T^* ,

$$\begin{aligned} \sum_{t=1}^T g_{i_T^*, t} &= \min_{i \in \{1, \dots, I\}} \sum_{t=1}^T g_{i, t} \\ &\leq \sum_{i=1}^I p_i^* \sum_{t=1}^T g_{i, t} \\ &= \sum_{t=1}^T \sum_{i=1}^I p_i^* R(d_i, \theta_t) \\ &= \sum_{t=1}^T R(p^*, \theta_t) \\ &\leq T \sup_{\theta \in \Theta} R(p^*, \theta) = T\bar{v}. \end{aligned} \quad (3)$$

Combining (2) and (3) gives

$$\begin{aligned} f(\bar{p}_T) - \bar{v} &\leq \frac{1}{T} \sum_{t=1}^T p_t^\top g_t - \bar{v} \\ &\leq \frac{1}{T} \sum_{t=1}^T (p_t^\top g_t - g_{i_T^*, t}). \end{aligned}$$

Applying (1) proves the result. $\square$

1.4 Comparison with the Hedge guarantee

Theorem 1 is a second-order, instance-dependent guarantee. Therefore, we must make some algebraic adjustments to make sense of it in terms of the same guarantees as for Hedge in Aradillas Fernández et al. (2025). Under Assumption A1 alone,

$$|p_t^\top g_t - g_{i_T^*, t}| \leq M,$$

and therefore

$$S_T^* \leq M^2 T.$$

Consequently, Theorem 1 always implies the worst-case bound

$$\begin{aligned} f(\bar{p}_T) - \bar{v} &\leq M \sqrt{\frac{2}{T}} \left(1 + \sqrt{2 \log \left[I \left(\frac{1}{2} + \log(T+1) \right) \right]} \right) \\ &\quad + \frac{5M}{T} \log[1 + I(1 + 2 \log(T+1))]. \end{aligned}$$

In order notation, this is

$$f(\bar{p}_T) - \bar{v} = O \left(M \sqrt{\frac{\log I + \log \log(T+1)}{T}} + \frac{M \{\log I + \log \log(T+1)\}}{T} \right).$$

Thus Squint does not improve the worst-case $T^{-1/2}$ convergence rate obtained by Hedge. In particular, the Hedge result

$$f(\bar{p}_T) - \bar{v} \leq M \sqrt{\frac{2 \log I}{T}}$$

has a slightly cleaner worst-case bound because Squint pays a logarithmic overhead for adapting to the realized second-order quantity. The benefit of Squint is instead that the leading term in Theorem 1 depends on S_T^* rather than its worst-case upper bound $M^2 T$. Whenever

$$S_T^* = o(M^2 T),$$

the second-order guarantee can therefore be strictly sharper than the corresponding worst-case Hedge guarantee.

References

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